

The Asymptotics of RMSEA and Its Robust Corrections

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Purpose

RMSEA is the sample root of a *population* quantity: the noncentrality of the model's discrepancy per degree of freedom. Its entire asymptotic meaning is fixed by one device, a sequence of local alternatives (Pitman drift), which turns the divergent exact-fit chi-square into a convergent *noncentral* chi-square whose noncentrality is what RMSEA estimates. The robust variants keep that target but replace the clean normal-theory limit by the general non-normal quadratic-form limit; the correction is exactly the removal of the mean inflation that non-normality injects into the noncentrality estimate.

This note records the chain twice. Under normality: exact-fit limit $\rightarrow$ local drift $\rightarrow$ noncentral $\chi^2 \rightarrow$ RMSEA. Under non-normality: general quadratic-form limit $\rightarrow$ mean inflation $\text{tr}(UT) \neq df \rightarrow$ scaled and robust RMSEA. The single load-bearing identity is that for a quadratic form the misspecification noncentrality and the sampling-noise mean are *additive and separable* (Theorem 3), so the robust point estimate is exact to leading order while only the reference distribution needs an approximation.

1 Setup and the exact-fit limit

Let $\hat{\sigma} = \text{vech } \hat{\Sigma}$ stack the $p^* = p(p+1)/2$ distinct sample covariances (append the means for a mean structure; the algebra is unchanged). The structural model is a smooth map $\theta \mapsto \sigma(\theta)$, $\theta \in \mathbb{R}^t$, with full-column-rank Jacobian $\dot{\sigma} = \partial\sigma/\partial\theta'$ ($p^* \times t$). The model has

$$df = p^* - t$$

degrees of freedom. A discrepancy $F(\hat{\sigma}; \sigma(\theta)) \geq 0$, minimized at $\hat{\theta}$, defines the test statistic

$$T = n\hat{F}, \quad \hat{F} = F\{\hat{\sigma}; \sigma(\hat{\theta})\},$$

where n is the sample-size multiplier: $n = N - 1$ in the classical Browne–Cudeck convention, $n = N$ in lavaan and in magmaan (where $\mathbf{fmin} = \frac{1}{2}F$ and $\mathbf{chi2_stat} = 2N \cdot \mathbf{fmin} = NF$). For ML, $F_{\text{ML}} = \text{tr}(\hat{\Sigma}\Sigma(\theta)^{-1}) - \log |\hat{\Sigma}\Sigma(\theta)^{-1}| - p$.

The local geometry is governed by the discrepancy’s Hessian in the moment argument at the truth, $\frac{1}{2}\partial^2 F/\partial\sigma\partial\sigma' = V$. For ML and GLS, $V = \frac{1}{2}D'(\Sigma^{-1} \otimes \Sigma^{-1})D$ with D the duplication matrix, and $V = \Gamma_{\text{NT}}^{-1}$, the inverse of the normal-theory covariance of $\sqrt{n}\hat{\sigma}$. Define the *residual-weight projector*

$$U = V - V\dot{\sigma}(\dot{\sigma}'V\dot{\sigma})^{-1}\dot{\sigma}'V, \quad U\dot{\sigma} = 0, \quad \text{rank } U = p^* - t = df. \quad (1)$$

U annihilates the model tangent space and keeps its V -orthogonal complement; it is the object every limit law below is a quadratic form in.

Theorem 1 (Exact-fit limit). *Suppose the model is correct, $\sigma_0 = \sigma(\theta_0)$, the data are i.i.d. multivariate normal, and θ_0 is identified with $\dot{\sigma}$ of full rank. Then $T \xrightarrow{d} \chi_{df}^2$.*

Proof. By the delta method $z := \sqrt{n}(\hat{\sigma} - \sigma_0) \xrightarrow{d} N(0, \Gamma)$ with $\Gamma = \Gamma_{\text{NT}} = V^{-1}$ under normality. A second-order expansion of $\hat{F}$ about the truth, profiled over θ , gives $T = z'Uz + o_p(1)$. With $\Gamma = V^{-1}$, the matrix $U\Gamma = UV^{-1}$ is idempotent: writing $P = \dot{\sigma}(\dot{\sigma}'V\dot{\sigma})^{-1}\dot{\sigma}'V$ (so $P^2 = P$), $UV^{-1} = I - VPV^{-1}$ and $(UV^{-1})^2 = UV^{-1}$, of rank $p^* - t = df$. A quadratic form $z'Uz$ with $z \sim N(0, \Gamma)$ and $U\Gamma$ idempotent of rank df is χ_{df}^2 . $\square$

So a correct model still spends $\mathbb{E}[T] \approx df$ of chi-square on pure sampling noise. The defect of T as a fit measure is the next line.

Corollary 1 (Why the exact-fit test is not a fit measure). *If the model is incorrect, with population discrepancy $F_0 = \min_{\theta} F\{\sigma_0; \sigma(\theta)\} > 0$, then $T = n\hat{F} \xrightarrow{p} \infty$ at rate n , so for any fixed nonzero misspecification the exact-fit test rejects with probability $\rightarrow 1$. The reject/retain dichotomy carries no information about the size of the misspecification at usable N .*

F_0 is the right population target: $F_0 = 0$ iff the model holds, and F_0 grades misspecification in the discrepancy metric. But $F_0 > 0$ fixed makes T diverge, so there is no limiting law to invert into a confidence statement. The resolution is to let the misspecification shrink with N .

2 Local asymptotics: the noncentral chi-square

The device, due to Steiger, Shapiro and Browne [11] following the general theory of sequential chi-square statistics and noncentral limits (Shapiro [8]; Satorra [5]), is a *sequence of local alternatives*: let the population move toward the model along a fixed off-model direction at the $\sqrt{n}$ rate,

$$\Sigma_N = \Sigma(\theta^*) + \Delta/\sqrt{n}, \quad \tau := \text{vech } \Delta \text{ fixed, } \tau \notin \text{col}(\dot{\sigma}). \quad (2)$$

Under (2) the population misfit is $O(1/n)$, calibrated so that $nF_0 \rightarrow \tau'U\tau < \infty$: misspecification and sampling noise become the same asymptotic order, which is the only regime in which the noncentrality stays finite in the limit.

Theorem 2 (Noncentral limit). *Under the drift (2), multivariate normality, and the regularity of Theorem 1,*

$$T \xrightarrow{d} \chi_{df}^2(\lambda), \quad \lambda = \tau'U\tau = \lim_N nF_0.$$

Proof. Under (2), $z = \sqrt{n}(\hat{\sigma} - \sigma(\theta^*)) \xrightarrow{d} N(\tau, \Gamma)$ with $\Gamma = V^{-1}$. As in Theorem 1, $T = z'Uz + o_p(1)$ with $U\Gamma$ idempotent of rank df . A quadratic form $z'Uz$ with $z \sim N(\tau, V^{-1})$ and UV^{-1} idempotent of rank df is noncentral $\chi_{df}^2(\lambda)$ with $\lambda = \tau'U\tau$ (the mean evaluated in the quadratic form). The same second-order expansion of the population discrepancy gives $nF_0 \rightarrow \tau'U\tau$, the identical quadratic form, because F_0 is read in the V -metric that defines U . $\square$

This is the asymptotic content of RMSEA. The limit $\mathbb{E}[T] \approx df + \lambda$ with $\lambda = nF_0$ separates the chi-square into a sampling part (df) and a misspecification part (λ). Estimate the latter by subtracting the former:

$$\hat{\lambda} = \max(T - df, 0), \quad (3)$$

the truncation enforcing $\lambda \geq 0$ against negative sampling realizations.

Remark (Definition of RMSEA). RMSEA is the root of the per-degree-of-freedom noncentrality on the discrepancy scale (Steiger–Lind [10]; Steiger [9]; Browne–Cudeck [3]),

$$\varepsilon = \sqrt{\frac{F_0}{df}} = \sqrt{\frac{\lambda}{n df}}, \quad \hat{\varepsilon} = \sqrt{\frac{\max(T - df, 0)}{n df}} = \sqrt{\max\left(\frac{\hat{F}}{df} - \frac{1}{n}, 0\right)}. \quad (4)$$

Dividing by df is the parsimony adjustment: among models with equal population misfit F_0 , the one spending more degrees of freedom (the simpler

model) earns the smaller ε . It is also what stabilizes the conventional cut-offs ($\varepsilon \lesssim 0.05$ close, 0.05–0.08 reasonable, $\gtrsim 0.10$ poor; Browne–Cudeck [3]) across models of different complexity, though those cutoffs are conventions, not asymptotic objects.

Corollary 2 (Confidence interval and tests of close fit). *Because $T \sim \chi_{df}^2(\lambda)$, a $(1 - \alpha)$ interval for λ is obtained by inverting the noncentral CDF: λ_L, λ_U solve $P\{\chi_{df}^2(\lambda_U) \leq T\} = \alpha/2$ and $P\{\chi_{df}^2(\lambda_L) \leq T\} = 1 - \alpha/2$, then map through (4). The test of close fit $H_0 : \varepsilon \leq \varepsilon^*$ (MacCallum, Browne and Sugawara [4]) and its power and sample-size planning are noncentral- χ^2 tail computations at $\lambda^* = n \, df \, \varepsilon^{*2}$. The interval, not the point estimate, is the reportable object: $\hat{\lambda}$ in (3) is biased and the truncation makes point RMSEA pile up at 0 for small true misfit.*

3 The non-normal limit and the mean inflation

Drop normality. The sample moments still satisfy $z = \sqrt{n}(\hat{\sigma} - \sigma_0) \xrightarrow{d} N(0, \Gamma)$, but now Γ is the general asymptotic covariance carrying fourth-order cumulants (Browne’s ADF matrix [2]), $\Gamma \neq V^{-1}$. The expansion $T = z'Uz + o_p(1)$ is unchanged, but $U\Gamma$ is no longer idempotent.

Lemma 1 (General quadratic-form limit). *Under the drift (2) with finite fourth moments,*

$$T \xrightarrow{d} \sum_{j=1}^{df} \alpha_j \chi_{1,j}^2(\delta_j),$$

a weighted sum of independent (noncentral) χ_1^2 variates, where $\alpha_1, \dots, \alpha_{df}$ are the nonzero eigenvalues of $U\Gamma$ and $\sum_j \alpha_j \delta_j = \tau'U\tau$. Under normality $\Gamma = V^{-1}$, all $\alpha_j = 1$, and the mixture collapses to $\chi_{df}^2(\tau'U\tau)$ of Theorem 2.

The estimator (3) silently assumed all $\alpha_j = 1$. The exact first moment shows what that costs.

Theorem 3 (Additive separation of the mean). *For any $z \sim (\mu, \Gamma)$ and symmetric U , $\mathbb{E}[z'Uz] = \text{tr}(U\Gamma) + \mu'U\mu$ exactly. Hence under the drift (2),*

$$\mathbb{E}[T] \rightarrow \underbrace{\text{tr}(U\Gamma)}_{\text{sampling, } = c \, df} + \underbrace{\tau'U\tau}_{\text{misspecification, } = nF_0}, \quad c := \frac{\text{tr}(U\Gamma)}{df}. \quad (5)$$

The scaling factor c is the Satorra–Bentler mean correction [6, 5]; $c = 1$ under normality and $c \neq 1$ (typically $c > 1$ under leptokurtosis) otherwise.

The misspecification term is the same quadratic form $\tau'U\tau = nF_0$ as in the normal case: it is built from U , hence from the discrepancy metric V , and does not depend on Γ .

Proof. The mean identity is the standard quadratic-form expectation. Apply it with $\mu = \tau$ and the general Γ ; the two terms are (5). The misspecification term is metric-invariant because F_0 is the population discrepancy in the V -metric, established in Theorem 2 independently of the sampling law Γ . $\square$

The consequence for the naive estimator is immediate and is the failure that the robust variants exist to fix.

Corollary 3 (The naive RMSEA is biased under non-normality). *Under non-normality, (3) satisfies $\mathbb{E}[T - df] \rightarrow df(c - 1) + nF_0$. The naive non-centrality estimate therefore carries a spurious term $df(c - 1)$ of sampling origin: when $c > 1$ it reports misspecification where there is none, and $\hat{\varepsilon}$ in (4) is inconsistent for ε .*

4 Robust RMSEA

Theorem 3 writes down its own correction: the sampling mean is $c df = \text{tr}(U\Gamma)$, not df , so subtract *that*. Estimate Γ by the sample fourth-moment (sandwich) matrix $\hat{\Gamma}$, set $\hat{c} = \text{tr}(U\hat{\Gamma})/df$, and define the noncentrality estimate and robust RMSEA

$$\hat{\lambda}_{\text{rob}} = \max(T - \hat{c} df, 0), \quad \hat{\varepsilon}_{\text{rob}} = \sqrt{\frac{\max(T - \hat{c} df, 0)}{n df}}. \quad (6)$$

This is the Brosseau-Liard–Savalei correction [1, 7], what lavaan reports as `rmsea.robust`. Its motivation is exactly (5): subtracting the *inflated* exact-fit mean $\hat{c} df$ removes the non-normality contamination and leaves an estimate of nF_0 alone.

Corollary 4 (Scaled vs. robust). *Let $T_{\text{sc}} = T/\hat{c}$ be the Satorra–Bentler scaled statistic and $\hat{\varepsilon}_{\text{sc}} = \sqrt{\max(T_{\text{sc}} - df, 0)/(n df)}$ the RMSEA naively recomputed from it (lavaan’s `rmsea.scaled`). Since $T - \hat{c} df = \hat{c}(T_{\text{sc}} - df)$,*

$$\hat{\varepsilon}_{\text{rob}} = \sqrt{\hat{c}} \hat{\varepsilon}_{\text{sc}}.$$

The two differ by $\sqrt{\hat{c}}$; with $\hat{c} > 1$ the robust value exceeds the scaled one. The scaled version divides the noncentrality estimate by $\hat{c}$ a second time (once in T_{sc} , never undone), which over-corrects; (6) is the version that targets ε consistently [7].

Remark (Point estimate exact, interval approximate). The robust *point* estimate rests only on the exact mean identity of Theorem 3, so it is correct to leading order regardless of the mixture weights α_j . The confidence interval is the soft spot: inverting a noncentral χ^2_{df} CDF (Corollary 2) approximates the true limit, the weighted mixture of Lemma 1, by a single noncentral χ^2 . The mean-and-variance (Satterthwaite) refinement additionally matches the second moment, replacing df by $df^* = \text{tr}(U\Gamma)^2 / \text{tr}\{(U\Gamma)^2\}$ and rescaling, so the reference $\chi^2_{df^*}$ has the right mean and variance; the robust RMSEA interval is then built on T rescaled to that reference. magmaan should expose the mean-scaled $\hat{c}$ for the point estimate and the df^* refinement for the interval, matching lavaan’s `rmsea.ci.*.robust`.

Remark (Asymptotic robustness: when $c = 1$ already). Non-normality does not always inflate. Under the asymptotic-robustness conditions (elliptical or independent-factor populations with the model class closed under the relevant transformations; Browne [2], Satorra [5]), $U\Gamma$ stays idempotent on the relevant subspace and $c \rightarrow 1$, so the naive RMSEA of (4) is already consistent and the robust correction is a no-op to leading order. The correction earns its keep precisely outside that class, where the fourth-cumulant structure couples to the residual space $\text{col}(U)$. This mirrors the FIML point-estimate story of the companion note: location robust, second-order objects (here the reference law, there the standard errors) fragile.

Remark (Categorical and multigroup). For categorical data fit by DWLS/ULS, the same algebra runs with $\hat{\sigma}$ the polychoric/threshold vector and Γ its asymptotic covariance; the diagonal weight matrix makes $c \neq 1$ generic, so the robust forms (6) (scaled, and scaled-and-shifted for the mean-and-variance reference) are the default, not an option. With G groups, $F_0 = \sum_g (N_g/N) F_{0,g}$ and RMSEA acquires the lavaan factor $\sqrt{G}$: $\hat{\varepsilon} = \sqrt{G} \sqrt{\max(T - df, 0)/(n df)}$, because the per-group noncentralities add while df counts the pooled residual dimension. The robust correction (6) composes with this multiplicatively through $\hat{c}$.

Remark (What RMSEA is not). RMSEA is a consistent estimate of ε , a population functional of the *discrepancy*, not of substantive misspecification directly; two models with the same F_0 but very different structural errors share an RMSEA (Yuan [12]). The local-alternative scaffolding is what licenses the noncentral- χ^2 machinery, but the reported ε is then read as a fixed population number, not as a statement that the truth is $O(1/\sqrt{n})$ from the model. The drift is a calculational device for a finite-misfit world, the same posture as power analysis under a fixed alternative.

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